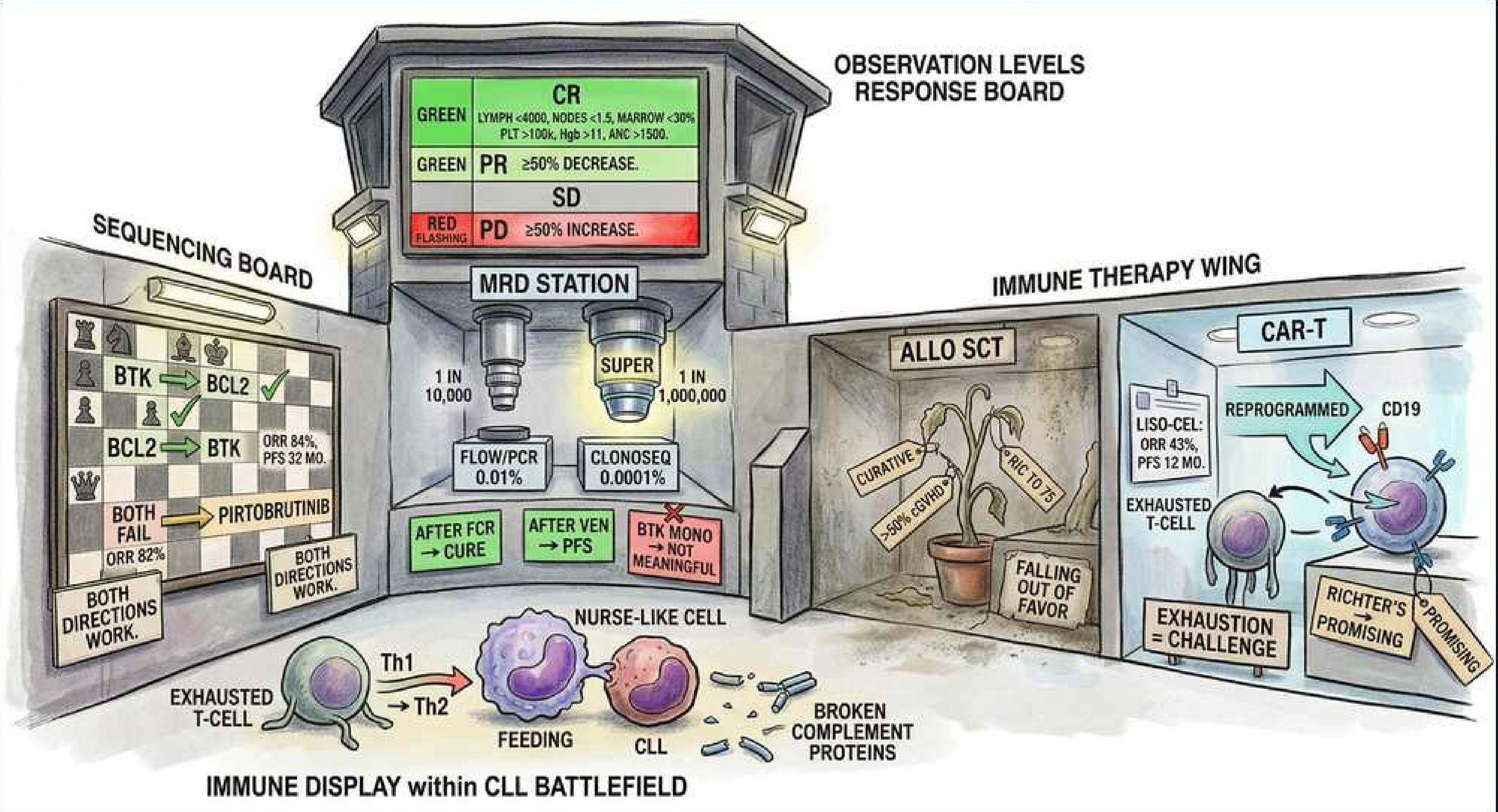
CLL — Response Assessment, MRD, Sequencing & Cellular Therapy



1. Response board

WHAT YOU SEE

A control-tower 'OBSERVATION LEVELS' response board, lit panels stacked CR / PR / SD / PD — the readout that grades how the disease answered therapy.

WHAT IT ENCODES

iwCLL response categories assessed ~2 months after therapy: CR (complete remission), PR (partial response, ≥50% drop), SD (stable disease), PD (progressive disease, ≥50% rise or new lesions). Response is graded on blood counts, nodes, spleen/liver size, and marrow — symptoms alone don't define response.

BOARD TRAP

Lymphocytosis can transiently SPIKE after starting a BTK inhibitor (redistribution of CLL cells out of nodes into blood) — this is NOT progressive disease and should not prompt stopping the drug.

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2. CR criteria

WHAT YOU SEE

The top GREEN bar 'CR' spelling out lymph <4000, nodes <1.5, marrow <30%, plt >100k, Hb >11 — green = the full, all-boxes-checked complete remission.

WHAT IT ENCODES

Complete remission requires ALL: blood lymphocytes <4×10□/L, no node >1.5 cm, no hepatosplenomegaly, no constitutional symptoms, marrow lymphocytes <30%, AND count recovery (platelets >100k, Hb >11 g/dL untransfused, neutrophils >1.5k). CRi = CR with incomplete marrow recovery.

BOARD TRAP

A normal-looking blood count is not CR — nodes, spleen, and marrow must all clear. CR with residual MRD still carries worse PFS than MRD-undetectable CR.

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3. PR ≥50%

WHAT YOU SEE

A second GREEN bar 'PR — ≥50% DECREASE' — green again but one tier down: a good-but-partial response.

WHAT IT ENCODES

Partial response = ≥50% reduction in blood lymphocyte count AND ≥50% reduction in nodal/organ burden, plus improvement in at least one cytopenia. PR-L (PR with lymphocytosis) is a recognized BTKi-specific category where nodes shrink but blood lymphocytes stay high due to redistribution.

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4. MRD 1 in 10,000

WHAT YOU SEE

The 'MRD STATION' with a standard microscope reading '1 in 10,000' — the conventional uMRD depth of 10□□ (0.01%) by flow.

WHAT IT ENCODES

Standard MRD-undetectable (uMRD) threshold = <1 CLL cell per 10,000 leukocytes = 10□□ = 0.01%, measured by 4–8 color flow cytometry or ASO-PCR. uMRD at 10□□ is the conventional cutoff correlating with prolonged PFS.

BOARD TRAP

"MRD-negative" is sensitivity-dependent — it means below the assay's detection limit, not zero disease. Always state the threshold (10□□ vs 10□□).

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5. Deep MRD 10^-6

WHAT YOU SEE

A larger 'SUPER' microscope reading '1 in 1,000,000 — clonoSEQ 0.0001%' — NGS that sees two logs deeper (10□□) than flow.

WHAT IT ENCODES

Next-generation sequencing (clonoSEQ, IGH-based) detects MRD down to 10□□ (1 in a million = 0.0001%), two logs deeper than flow. Deeper uMRD predicts more durable remission and is increasingly used as a trial endpoint.

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6. MRD after venetoclax

WHAT YOU SEE

A GREEN box 'AFTER VEN → PFS' — green = MRD is meaningful and useful after fixed-duration venetoclax.

WHAT IT ENCODES

MRD is meaningful with FIXED-DURATION venetoclax-based regimens (e.g., venetoclax + obinutuzumab): achieving uMRD predicts longer PFS and supports time-limited therapy. MRD-guided trials use uMRD to decide when to stop venetoclax.

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7. MRD not for BTK

WHAT YOU SEE

A RED box 'BTK MONO → NOT MEANINGFUL' — red flags that chasing MRD on continuous BTKi is pointless; it never clears.

WHAT IT ENCODES

MRD is NOT meaningful for continuous BTK-inhibitor monotherapy: BTKi produces durable disease CONTROL without deep clearance, so MRD stays detectable indefinitely even in excellent responders. Treatment is continued until progression or toxicity, not until uMRD.

BOARD TRAP

Don't use MRD to decide when to stop a BTK inhibitor — uMRD is rarely reached and is not the goal of continuous therapy. MRD-guided stopping applies to venetoclax regimens.

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8. Sequencing: both work

WHAT YOU SEE

A chessboard 'SEQUENCING BOARD' with green arrows BTK→BCL2 and BCL2→BTK both checkmarked 'both directions work' — either order is a winning move.

WHAT IT ENCODES

BTK inhibitor and BCL2 inhibitor (venetoclax) can be sequenced in EITHER order after failure of the first — both directions retain efficacy. Choice of first agent depends on patient factors (cardiac/bleeding risk favors avoiding ibrutinib; TLS risk and adherence to ramp-up affect venetoclax).

9. Pirtobrutinib after both

WHAT YOU SEE

A yellow tag 'PIRTOBRUTINIB — both fail → ORR 82%' at the bottom of the chessboard — the reversible BTKi that still scores after both prior pieces are taken.

WHAT IT ENCODES

Pirtobrutinib is a NONCOVALENT (reversible) BTK inhibitor that binds independent of the C481 cysteine, so it salvages disease resistant to covalent BTKi via the C481S mutation, and works even after both covalent BTKi and venetoclax failure (ORR ~82% in the double-exposed BRUIN data).

BOARD TRAP

C481S mutation = classic resistance mechanism to COVALENT BTKi (ibrutinib/acalabrutinib/zanubrutinib); pirtobrutinib overcomes it because it doesn't need that cysteine.

10. Allo SCT

WHAT YOU SEE

The 'ALLO SCT' room with a wilting potted plant, tags 'CURATIVE' but 'FALLING OUT OF FAVOR' — the only curative but fading option.

WHAT IT ENCODES

Allogeneic stem cell transplant is the only potentially CURATIVE option (graft-versus-leukemia effect) but carries high transplant-related mortality. It has fallen out of favor, now reserved for young, fit, double-refractory (post-BTKi and post-venetoclax) or high-risk patients — largely supplanted by targeted agents and CAR-T.

11. CAR-T CD19

WHAT YOU SEE

The glowing 'CAR-T' chamber: a T-cell being 'REPROGRAMMED' with an arrow to 'CD19' on the target B-cell (liso-cel, ORR 43%).

WHAT IT ENCODES

Anti-CD19 CAR-T (liso-cel, lisocabtagene maraleucel) is approved for relapsed/refractory CLL/SLL after ≥2 prior lines including a BTKi and venetoclax. Autologous T cells are reprogrammed to target CD19 on the malignant B cells.

BOARD TRAP

Watch for CAR-T toxicities: cytokine release syndrome (CRS, treat with tocilizumab) and ICANS (neurotoxicity).

12. T-cell exhaustion

WHAT YOU SEE

A drooping 'EXHAUSTED T-CELL' tagged 'EXHAUSTION = CHALLENGE' — the worn-out T cell that is CLL's central barrier to cell therapy.

WHAT IT ENCODES

CLL induces profound T-cell exhaustion (upregulated PD-1, impaired synapse formation), which historically limited CAR-T efficacy compared to other B-cell malignancies — a central biologic challenge for cellular therapy in CLL.

13. CAR-T for Richter

WHAT YOU SEE

A tag reading 'RICHTER'S — PROMISING' pinned in the CAR-T wing — the one transformation setting where cell therapy looks hopeful.

WHAT IT ENCODES

CAR-T (and allo-SCT) show promise for Richter transformation (DLBCL evolving from CLL), which otherwise responds poorly to standard DLBCL chemoimmunotherapy and carries a dismal prognosis.

14. Nurse-like cells

WHAT YOU SEE

A purple 'NURSE-LIKE CELL' literally 'FEEDING' a CLL cell, with Th1→Th2 arrow — the stromal nanny that shelters CLL in the node.

WHAT IT ENCODES

The tumor microenvironment sustains CLL: nurse-like cells (monocyte-derived) feed and protect CLL cells via CD40L/BAFF/APRIL and chemokines, Th1/Th2 skewing drives immune dysfunction, and broken/consumed complement proteins contribute to infection risk. This stromal dependence underlies sanctuary in lymph nodes.

15. Broken complement

WHAT YOU SEE

A scattered pile of snapped Y-shaped 'BROKEN COMPLEMENT PROTEINS' — the shattered immune defenses that make infection the top killer.

WHAT IT ENCODES

Complement deficiency/dysfunction plus hypogammaglobulinemia and T/NK dysfunction together produce the multi-arm immune failure of CLL, explaining why infection is the leading cause of death even before cytotoxic therapy.